F Additional Empirical Results

F.1 Impact of Different Segments of the Prompt in Every Layer

In Section 6, we elucidate how comparing the norm of each VJP (δ_i) involved in the construction of a gradient matrix can reflect which layers are updated more than others. Similar comparison is conducted for the tokens from the edited prompt, uncovering that certain layers and segmentation from the prompt do not contribute significantly to the updating process.

We extend this analysis from Section 6 to demonstrate the ability to identify the main editing matrices in every type of module in LMs Figure 16.

We will solely discuss the results related to the MLP layers, as we did not examine the attention modules in our study. Both the FF_1 module (*mlp.c_fc*) and FF_2 module (*mlp.c_proj*) demonstrate that the primary editing occurs around the first quarter of layers by the last subject token, and around the middle of the layers by the last token. The majority of the other layers and tokens exhibit VJP norms close to zero, indicating that they scarcely contribute to updating the model's weights. The same pattern is also evident for the VJPs between transformer layers (*transformer.h*), as, if we disregard Dropouts and Layer Norms, the VJPs of FF_2 are the same as the one between the transformer layers.

We also provide similar figures where, instead of plotting the norm of the VJPs, we compare the norms of the intermediate inputs to every layer (x_i) (Figure 17). The inclusion of those figures is solely to emphasize that there is no correlation between the norm of x_i and δ_i.

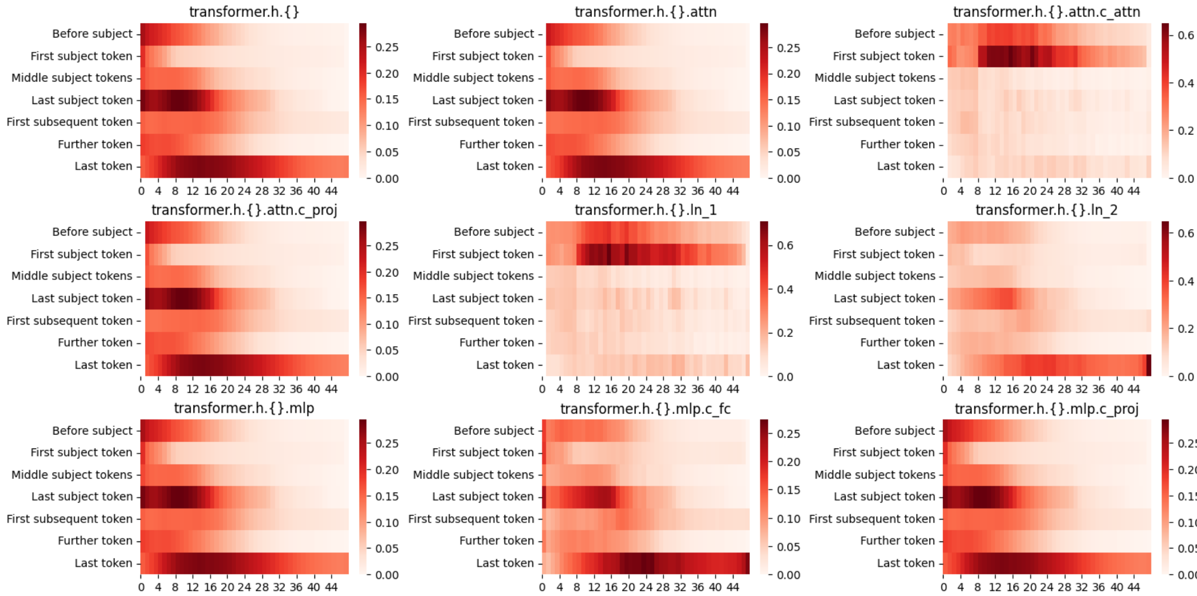

Figure 16: δ_i norm for each of GPT2-xl sub-module (names are according to the HuggingFace implementation). This is an extension to Section 6, showing how our approach is easily adapted to any submodule of the model. The layer of FF_1 is represented by *mlp.c_fc* and FF_2 by *mlp.c_proj* .

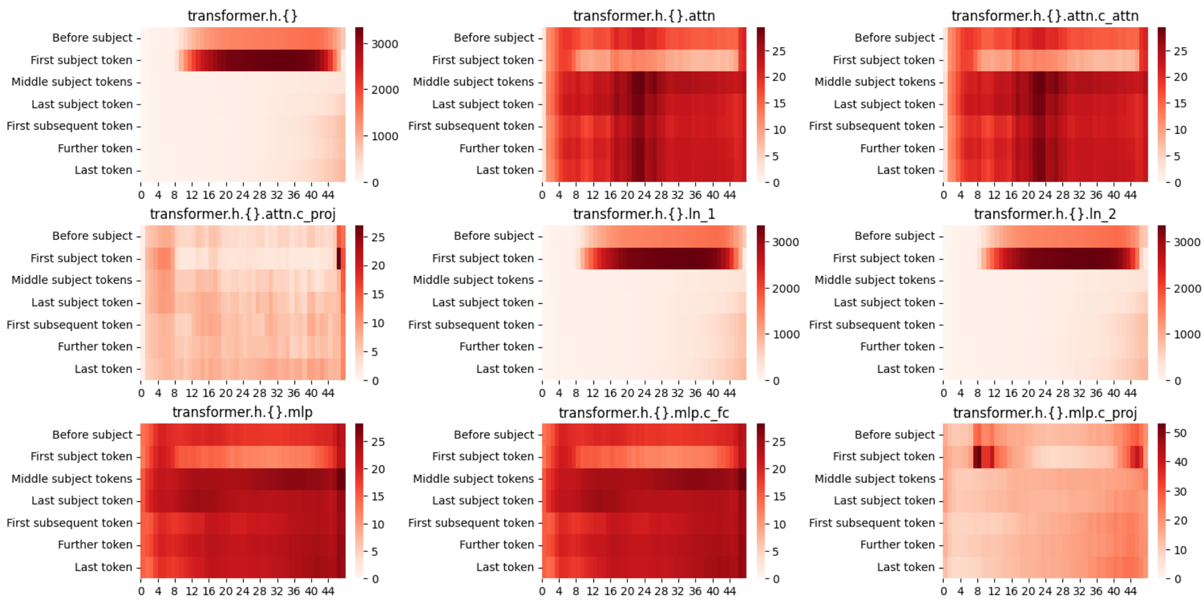

Figure 17: x_i norm for GPT2-xl. Provided to show no correlation with Figure 16.

F.2 The Ranks of FF_1 and the Models' Original Answer

In Section 6 we examine how FF_2's VJPs rank the target tokens, the tokens we try to learn during backpropagation. One possible interpretation of our analysis is to examine the backward pass according to the gradual change in the embeddings it tries to inject (add or subtract) into the model. Previous works conduct similar analysis with the hidden states from the forward pass Haviv et al. (2023); Geva et al. (2022b). Those works examined the gradual build in LMs' forward pass prediction, from the perspective of the last token in the edited prompts. In this section, we expand upon these examinations by measuring the LL rank for the original answers outputted by the forward pass (before editing), and by observing these LL ranks from the perspective of FF_1's spanning set.

The presented results are based on 100 distinct edits using a single backward pass per edit. We employ GPT2 and Llama2-7B, and the edited prompts and targets were taken from the CounterFact dataset.

FF_2 : In Section 6 we presented the ranking of the target token for GPT2-xl. Subsequently, we present the rank of the last prompt's token VJP (annotated with δ_n in Lemma 5.1) also for GPT2-medium and Llama-7B. In Figure 18 we can see that all models' LL projections rank the target token as one of the most improbable tokens along most of the layers, with some degradation in the first few layers. We associate this drop to the gap of LL in projection vectors from earlier layers.

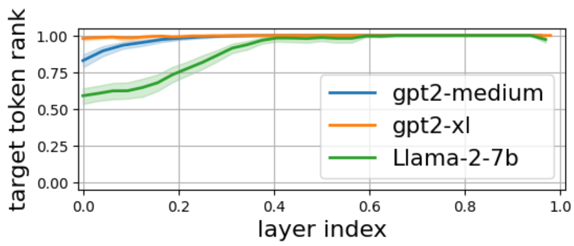

Figure 18: The Logit Lens rank of the target token in VJP of the last token from the edited prompt, δ_n. In order to show different models in the same figure, we normalize the layer indices and the rank of the token in the vocabularies (which is 50K for GPT2 and 32K for Llama2). All models' δ_n assign the target token as one of the most improbable tokens along most of their layers.

We repeat the same experiment and measure the rank of the original token predicted by the model. According to Equation 4 and Lemma 5.1, the embedding of a token in the gradient should be discernible if it is the target token or if its probability in the model's final output is relatively high.

The results depicted in Figure 19 illustrate that the LL rank of the final answer is relatively low in the model's last layers, although not at the lowest possible level.

In Section 4.2 and Section 5.2, we delved into how the injection of the gradient vector with a high LL rank of the target token implies that the update aims to enhance the target probability in the output. Similarly, the observed pattern regarding the LL rank of the model's final answer suggests that the updates attempt to diminish the probabilities of the final answers in the model's output, but in a much smoother manner than those of the target token.

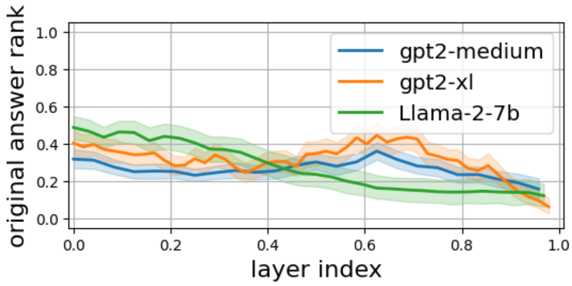

Figure 19: The Logit Lens ranks for the models' actual answers according to FF_2 gradients of the last prompts' token (layer indices and ranks are normalized). In the later layers, the rank of the original answers suggests that the model subtracts the embedding of those tokens from the model's weights.

FF_1 : Since this layer's δ_i is not projectable via LL, we use its inputs x_i as its spanning set. Analyzing the ranks of x_i is almost identical to the analysis of Haviv et al. (2023). We share our analysis in Figure 20, mainly to emphasize that gradients write into FF_1's weights the inputs x_i from the forward pass. The gradual build in the models' predictions becomes more apparent when we filter out examples where the model answers CounterFact's prompts incorrectly, accounting for approximately 84% of the instances with GPT2-medium/xl and Llama.[1][2] In Figure 20, we also included the same analysis with 50 correctly answered prompts.

[1]Most of the time, they predict tokens like "a", "is", and "the", rather than factual notions in the context of the prompts.

[2]Llama utilizes two matrices that employ FF_1 as part of its implementation of SwiGLU as its MLP activation function. Notably, the input x_i remains consistent for both matrices.